



DEFINITE INTEGRALS AND AREAS
DAY- 06 (21-03-2024)

SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

1. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$; $n \in N$, then which of the following statements hold good?

A) $2nI_{n+1} = 2^{-n} + (2n-1)I_n$

B) $I_2 = \frac{\pi}{8} + \frac{1}{4}$

C) $I_2 = \frac{\pi}{8} - \frac{1}{4}$

D) $I_3 = \frac{\pi}{16} - \frac{5}{48}$

2. Let $f(x) = \int_0^x |2t-3| dt$, then 'f' is

A) Continuous at $x = 3/2$

B) continuous at $x = 3$

C) Differentiable at $x = 3/2$

D) differentiable at $x = 0$

3. The value(s) of $\int_{-\infty}^0 \frac{e^{3x}(1-e^x)^4}{1+e^{-2x}} dx$ is/are

A) $\frac{22}{7} - \pi$

B) $\frac{2}{105}$

C) 0

D) $\log_e e - \int_1^2 [x] dx$ where $[]$ G.I.F.

4. $\int_{\frac{1}{\sqrt{3}}}^{\frac{\sqrt{3}}{2}} \frac{dt}{\left(1+t^{\frac{3}{2}}\right)(1+t^2)}$ Equal to $\frac{\pi}{p}$ then

A) The number of positive integral divisors of p is 6

B) The number of even positive integral divisors of p is 4

C) The sum of positive integral divisors of p is 28

D) The sum of even positive integral divisors of p is 24

5. Which of the following statements is/are false?

A) $\int_1^4 e^{x^2} d(x^2) = e^{16} - e^1$

B) If $0 < x < \frac{\pi}{2}$, $\int_{1/2}^{\sqrt{3}/2} \ln(e^{\cos x}) d(\sin x) = \frac{\pi}{12}$

C) Let $f(x) = \int_0^x |\cos t| dt$. Then $f'(x)$ is continuous but not differentiable at $x = \frac{\pi}{2}$

D) Let $f(x) = \int_{-2}^x |t+1| dt$, $x \in \left[-\frac{3}{2}, 1\right]$. Then, f is continuous but not differentiable at $x = -1$

6. Which of the following statements is/are true?

A) Let f be strictly increasing continuous function defined on the set of real numbers. If $f(1) = 2$, $f(2) = 4$ then $\int_1^2 f(x) dx$ can be equal to $\frac{25}{8}$

B) Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is such that $f''(x) \geq 0, \forall x \in [a, b]$. Then the value of $\int_a^b f(x) dx$ can not exceed $\frac{(f(a) + f(b))(b-a)}{2}$

C) The value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} < \frac{\pi}{4}$

D) the value of $\int_0^1 \frac{dx}{1+x^{\pi/2}} > \ln 2$

7. Let $f: \mathbb{R} \rightarrow (0, \infty)$ be a real valued function such that $\int_0^x t f(x-t) dt = e^{2x} - 1$, then which of the following is/are correct?

A) $f(0) = 4$ B) $(f^{-1})^1(4) = \frac{1}{8}$ C) $\lim_{x \rightarrow 0} \frac{f(x) - 4}{x} = 4$ D) $\left(\frac{df}{d(e^x)} \right)_{at x=0} = 8$

8. If $p = \frac{1}{n^4} \prod_{r=1}^{2n} (n^2 + r^2)^{\frac{1}{n}}$ then $\lim_{n \rightarrow \infty} \ln P$ is equal to

A) $\int_0^2 \ln(1+x^2) dx$

B) $\int_0^1 \ln(x^2 - 4x + 5) dx$

C) $\int_0^2 \ln(x^2 - 4x + 5) dx$

D) $2 \int_0^1 \ln(1+x^2) dx$

9. Let $I_1 = \int_0^{\pi/4} e^{x^2} dx$, $I_2 = \int_0^{\pi/4} e^x dx$, $I_3 = \int_0^{\pi/4} e^{x^2} \cos x dx$, $I_4 = \int_0^{\pi/4} e^{x^2} \sin x dx$ then which of the following is/are not TRUE?

A) $I_2 > I_4 > I_3$

B) $I_2 \geq I_1 \geq I_3$

C) $I_1 > I_3 > I_4$

D) $I_1 \geq I_3 \geq I_4$

10. $I_{m,n} = \int_0^1 x^m (\ln x)^n dx =$

A) $-\frac{m+1}{n+1} I_{m,n+1}$

B) $-\frac{m}{n+1} I_{m,n-1}$

C) $-\frac{n}{m+1} I_{m,n-1}$

D) $\frac{m}{n+1} I_{m,n-1}$

11. Which of the following inequalities is/are not INCORRECT?

A) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

B) $2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2$

C) $\frac{1}{3} < \int_0^1 x^{(\sin x + \cos x)^2} dx < \frac{1}{2}$

D) $\frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{5}{6}$

12. Let $S = \int_1^{e^2} \frac{\ln x}{\sqrt{x}} dx$ then which of the following is/are correct?

A) $S < 2\left(e - \frac{1}{e}\right)$

B) $S < (e-1)\left(2 + \frac{1}{\sqrt{e}}\right)$

C) $S = \int_0^2 t\sqrt{e^t} dt$

D) $3 < S < 5$

13. If $I = \int_0^{\pi/2} e^{-\lambda \sin x} dx$, $\lambda \in (0, \infty)$ then which of the following is/are true?

A) $I < \frac{\pi}{2}$

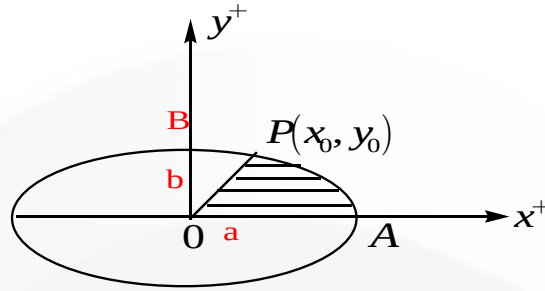
B) $I > \frac{\pi}{2}(e^{-\lambda} + 1)$

C) $I > \frac{\pi}{2}e^{-\lambda}$

D) $I > 0$

14. Let $f = [0, \infty) \rightarrow \mathbb{R}$ be continuously strictly increasing function such that $f^3(x) = \int_0^x t f^2(t) dt$ for every $x \geq 0$, then which of the following is/are correct?
- A) $f(6) = 6$
- B) $f(x)$ is onto
- C) $\int_0^1 f(x) dx = \frac{1}{18}$
- D) Number of solution of equation $f(x) = 6$ are two
15. If is the area bounded by $|x - a_i| + |y| = b_i, i \in \mathbb{N}$ and $a_{i+1} = a_i + \frac{3}{2}b_i$ and $b_{i+1} = \frac{b_i}{2}, a_1 = 0, b_1 = 32$ then
- A) $A_3 = 128$ B) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{4}{3}(16)^2$ C) $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{8}{3}(32)^2$ D) $A_3 = 256$
16. Which of the following statement(s) is/are true
- A) The area bounded by the curves $y = -\sqrt{-x}$ and $x = -\sqrt{-y}, x, y \leq 0$ is $1/3$
- B) The area bounded by the curves $y = \sqrt{-x}$ and $x = \sqrt{-y}, x, y \leq 0$ is $1/3$
- C) The area bounded by the curves $y = \sqrt{-x}$ and $x = \sqrt{-y}, x, y \leq 0$ and $y = 1$ is $2/3$
- D) The area bounded by the curves $y = x^2$ and $y = \sqrt{|x|}$, is $2/3$
17. The ratio of the area in which the curve $y = \left[\frac{x^3}{100} + \frac{x}{35} \right]$ divided the circle $x^2 + y^2 - 4x + 2y + 1 = 0$, where $[.]$ denotes the greatest integer function is
- A) $4\pi - 3\sqrt{3} : 8\pi + 3\sqrt{3}$ B) $4\pi + 3\sqrt{3} : 8\pi + 3\sqrt{3}$
- C) $4\pi\sqrt{3} - 9 : 8\pi\sqrt{3} + 9$ D) $4\pi - 3\sqrt{3} : 8\pi - 2\sqrt{3}$
18. Area of the region enclosed by the curves $C_1 : y = \operatorname{sgn} x, C_2 : x = 0, C_3 : y = x$ is not equal to
- A) 1 B) 2 C) $\frac{1}{2}$ D) 0

19. An ellipse shown here has equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$. If $P(x_0, y_0)$ is any point on it and area of curvilinear part OPA is denoted by Δ_0 . Then which of the following statements can be wrong?



- A) $P = \left(a \cos \left(\frac{2\Delta_0}{ab} \right), b \sin \left(\frac{2\Delta_0}{ab} \right) \right)$ B) $P = \left(a \cos \left(\frac{2\sqrt{\Delta_0}}{ab} \right), b \sin \left(\frac{2\sqrt{\Delta_0}}{ab} \right) \right)$
- C) $P = \left(a \cos \left(\frac{ab}{2\Delta_0} \right), b \sin \left(\frac{ab}{2\Delta_0} \right) \right)$ D) $P = \left(a \cos \left(\frac{\sqrt{ab}}{2\Delta_0} \right), b \sin \left(\frac{\sqrt{ab}}{2\Delta_0} \right) \right)$
20. Let $f(x)$ and $g(x)$ be two continuous functions on $a \leq x \leq b$ intersecting at only one point in (a, b) and let $p(x) = \max. \{f(x), g(x)\}$ and $q(x) = \min \{f(x), g(x)\}$ then the area bounded by the curves $y = p(x)$, $y = q(x)$ and ordinates $x = a$ and $x = b$ is given by
- A) $\int_a^b (f(x) - g(x)) dx$ B) $\int_a^b (p(x) - q(x)) dx$
- C) $\int_a^b |f(x) - g(x)| dx$ D) $\int_a^b |p(x) - q(x)| dx$
21. Five regions are defined by
 $C_1 : |x + y| \leq 1$, $C_2 : |x - y| \leq 1$, $C_3 : |x| \leq \frac{1}{2}$, $C_4 : |y| \leq \frac{1}{2}$
 $C_5 : 3x^2 + 3y^2 = 1$: then
- A) The area bounded by C_1, C_2 which doesn't contain area of C_5 is $2 - \frac{\pi}{3}$
- B) The part of area of C_5 which doesn't contain area of C_3 and C_4 is $\frac{2\pi}{9} - \frac{1}{\sqrt{3}}$
- C) The part of area bounded by C_1, C_2 but not bounded by C_3 and C_4 is 1
- D) Area bounded by C_1, C_2, C_3, C_5 but not by C_4 is $\frac{\pi}{2} - \frac{1}{\sqrt{3}}$
22. If $A\left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $B\left(-\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $C\left(-\frac{3}{\sqrt{2}}, -\sqrt{2}\right)$ and $D(3 \cos \theta, 2 \sin \theta)$ are four

Points, then the value of θ for which the area of quadrilateral ABCD is

Maximum, $\left(\frac{3\pi}{2} \leq \theta \leq 2\pi\right)$ is (in sq. units) is k , where $[k]$ is greater than or equal to

([.] Denotes G.I.F)

- A) 2 B) 3 C) 4 D) 5

23. Let $g^1(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in R$ with $g\left(\frac{1}{2}\right) = 0$. If A be the area of the region bounded by $y = g(x)$, with x -axis and between the ordinates $x = \frac{1}{2}$ and $x = 1$ such that $\alpha \leq A \leq \beta$, then the possible values of α and β are

- A) $\alpha = 14, \beta = 26$ B) $\alpha = \frac{1}{12}, \beta = \frac{1}{2}$
C) $\alpha = -12, \beta = 0$ D) $\alpha = 1, \beta = 12$

24. The area bounded by $y = \frac{\sin x}{x}$, x -axis and the ordinates $x = 0, x = \frac{\pi}{4}$ is

- A) ≤ 1 B) $< \frac{\pi}{4}$ C) $\leq \frac{\pi}{2}$ D) $\leq \frac{\pi}{3}$

25. Let $f(x) = \frac{x}{1+x^2}$ and $g(x) = \frac{1}{x(1+x^2)}$. If A_1 be the area of region bounded between

$y = f(x)$, x -axis, $x = \frac{1}{e}$ and $x = \pi$, and A_2 is the area of region bounded between $y = g(x)$,

x -axis and between $x = \frac{1}{\pi}$ and $x = \frac{1}{e}$. Then $|A_1 - A_2|$ is greater than or equal to

- A) 2 B) 1 C) $\ln \frac{3}{2}$ D) $\ln \frac{13}{11}$

SECTION – II

(INTEGER ANSWER TYPE)

1. Evaluate: $\int_{1/2}^2 \frac{1}{x} \sin\left(x - \frac{1}{x}\right) dx$

2. The value of $\int_0^1 \frac{1}{(5+2x-2x^2)(1+e^{(2-4x)})} dx$ is $\frac{1}{2\sqrt{a}} \ln \frac{\sqrt{a}+b}{\sqrt{a}-b}$ then $a-2b =$
3. If $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$, then $\frac{8}{\pi} \int_0^\infty \frac{\sin^3 x}{x} dx =$
4. If $\int_0^{2\pi} \frac{dx}{2+\sin 2x} = k \int_0^{\pi/2} \frac{dt}{7+\cos 2t}$, then the value of $\frac{k}{4}$ is _____
5. Let $a \in \left(0, \frac{\pi}{2}\right)$, then the value of $\lim_{a \rightarrow 0} \frac{1}{a^3} \int_0^a \ln(1+\tan a \tan x) dx$ is equal to $\frac{1}{k}$ where k is ____
6. Let $f(x)$ be a periodic function with fundamental period 12. If $\int_{-6}^{12} f(x) dx = 9$, $\int_0^{12} f(x) dx = 12$ then the value of $\frac{1}{3} \left| \int_0^6 f(x) dx \right|$ is _____
7. If $\lim_{n \rightarrow \infty} n^2 \int_{-1/n}^{1/n} (2015 \sin x + 2016 \cos x) |x| dx = \lambda$, then the sum of digits of λ is ____
8. $\int_0^\pi \sqrt{(\cos x + \cos 2x + \cos 3x)^2 + (\sin x + \sin 2x + \sin 3x)^2} dx$ has the value equal to $\frac{\pi}{k} + \sqrt{w}$, where k and w are positive integers. The sum of digits of HCF of k^w and w^k is
9. Let a, b, c be positive constants. When the value of integral $\int_0^1 (acx^{b+1} + a^3bx^{3b+5}) dx$ is independent of b , then $a^p c = \frac{r}{q}$, where p, q, r are expressed in simplest form, then $p+q+r =$
10. Let $f(x)$ be a continuous function defined from $[0, 2] \rightarrow R$ and satisfying the equation $\int_0^2 f(x)(x-f(x)) dx = \frac{2}{3}$ then the value of $2f(1) =$ _____

11. Let $A_n = \int_{\frac{1}{n+1}}^{\frac{n}{n+1}} \frac{\tan^{-1} nx}{\sin^{-1} nx} dx$ and $\lim_{n \rightarrow \infty} n^2 A_n = L$ then $8L =$
12. Consider a rectangle ABCD formed by the points A(0,0) B(6,0) C(6,4) D(0,4).
P is a moving interior point of the rectangle, moving in such a way $d(P, AB) \leq \min\{d(P, BC), d(P, CD), d(P, AD)\}$ and here $d(P, AB), d(P, BC), d(P, CD), d(P, AD)$ Represents that the distances of the point P from the sides AB, BC, CD and AD respectively. Area of the region representing all possible positions of the point P is equal to.....
13. Consider $f(x) = \begin{cases} \cos x, & 0 \leq x < \frac{\pi}{2} \\ \left(\frac{\pi}{2} - x\right)^2, & \frac{\pi}{2} \leq x < \pi \end{cases}$ such that f(x) is periodic with period π then the area bounded by $y = f(x)$ and the x-axis from $x = -n\pi$ to $x = n\pi$ is $an\left(1 + \frac{\pi^3}{b}\right)$, $n \in N$ for a given n, n is a natural number then $b/(2a) = \dots\dots$
14. Let $f(x) = x^3 + 3x + 2$ and $g(x)$ is the inverse of it. The area bounded by $g(x)$, the x-axis and the ordinates at $x = -2$ and $x = 6$ is $\frac{m}{n}$ where $m, n \in N$ & G.C.D of $(m, n) = 1$ then $m - 2 =$
15. If the area bounded by the curve $(y - \arcsin x)^2 = x - x^2$, is 'A', then the value of $[4A]$ is equal to ([.] denotes G.I.F)
16. Let $f(x) = x^2 + 6x + 1$ and S denotes the set of points (x, y) in the coordinate plane such that $f(x) + f(y) \leq 0$ and $f(x) - f(y) \leq 0$ the area of S is $k\pi$ then $k = \dots\dots$
17. If the area bounded by $y = \left[\sin^2 \frac{x}{2} + \sqrt{\cos \frac{x}{2}}\right]$ and $y = \left|\sin x - \frac{1}{2}\right|$, $x = 0, x = \pi$ is 'S', then $[2S] = \dots\dots$ (Where [.] represents the greatest integer function)
18. The area of the region bounded by the curves $y = \frac{4 - x^2}{4 + x^2}$, $25y^2 = 9x$ and $y = \frac{3}{5}|x| - \frac{6}{5}$ which contains (1, 0) point in its interior, is $\pi - a \tan^{-1} \frac{1}{2} + \frac{1}{b}$ where $b - 2a$ is equal to
(a, b are positive integers)
19. The minimum area bounded by the function $y = f(x)$ and $y = \alpha x + 9$ ($\alpha \in R$)

Yashpatil TG~ @bohring_bot

Where f satisfies the relation $f(x+y) = f(x) + f(y) + y\sqrt{f(x)} \quad \forall x, y \in \mathbb{R}$ and

$f'(0) = 0$ is 9A, value of 'A' is

20. If the area bounded by the region $|y|+1 = 2e^{-|x|}$ is $k(1-\log_e 2)$ then the numerical Value of 'k' is
21. Area of the region enclosed by all the curves $x^2 + y^2 = 4$, $y = x^2 + x + 1$, $y = \left[\sin^2 \frac{x}{4} + \cos \frac{x}{4} \right]$ and x-axis (where $[.]$ denotes G.I.F) is $\frac{K\pi}{6} + \frac{A\sqrt{3}}{2} - \frac{1}{B}$ where $(K, A, B \in \mathbb{N})$. Then the value of $\left[K + A + \frac{B}{2} \right]$ equals to
22. Let C be the curve passing through the point (1,1) has the property that the Perpendicular distance of the origin from the normal at any point P of the curve is Equal to the distance of P from the x-axis, If the area bounded by the curve C and x - Axis in the first quadrant is $\frac{k\pi}{2}$ square units, then the value of 'k' is equal to
23. If A is the area between the curve $y = x^6(1-x)^7$ and x - axis then the number of prime positive integral factors of A^{-1} is ____
24. Let the straight line $y = k$ divide the area enclosed by $x = (1-y)^2$, $x = 0$ and $y = 0$ into two parts of areas, $A_1 (0 \leq y \leq k)$ and $A_2 (k \leq y \leq 1)$ such that $\frac{A_1}{A_2} = 7$. Then $\frac{1}{k}$ is equal to
25. Let $f(x) = \int_x^{x^2 + \frac{\pi}{3}} 4\sin^2 t \, dt$ for all $x \in \mathbb{R}$ and $g: \left[0, \frac{1}{2}\right] \rightarrow [0, \infty]$ be a continuous Function. For $\alpha \in \left[0, \frac{1}{2}\right]$, if $f^1(\alpha) + 2$ is the area of the region bounded by $x = 0, y = 0, y = g(x)$ And $x = \alpha$, then the value of $g(0)$ is equal to
26. If A be the area of the region represented by $\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\}$, Then the value of 4A is equal to

(PARAGRAPH TYPE)

Paragraph-1

Let $g(t) = \int_{x_1}^{x_2} f(t, x) dx$. Then $g'(t) = \int_{x_1}^{x_2} \frac{\partial}{\partial t} (f(t, x)) dx$. Consider $f(x) = \int_0^\pi \frac{\ln(1 + x \cos \theta)}{\cos \theta} d\theta$

1. Range of $f(x)$ is

- A) $(0, \pi)$ B) $(0, \pi^2)$ C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ D) $\left(-\frac{\pi^2}{2}, \frac{\pi^2}{2}\right)$

2. $f(x)$ is

- A) Discontinuous at $x = 0$ B) continuous but not differentiable at $x = \frac{1}{2}$
C) Continuous at $x = 0$ D) not differentiable at $x = \frac{3}{4}$

Paragraph-2

Let f be a function defined by $f(k) = \int_0^{\pi/2} |k - 2 \sin t| \cos t dt$ where k is real parameter. Then

3. Which of the following statements is correct?

- A) $f(k)$ is continuous $\forall k$ but not derivable at two values of k .
B) $f(k)$ is differentiable $\forall k$
C) $f(k) = 1 - k \forall k \geq 2$
D) $f(k) = k - 1 \forall k \geq 0$

4. The minimum value of $f(k)$ is

- A) $\frac{1}{2}$ B) $\frac{3}{2}$ C) 0 D) 1

Paragraph-3

$$\text{Let } f_n(x) = \sum_{r=1}^n \frac{\sin^2 x}{\cos^2\left(\frac{x}{2}\right) - \cos^2\left(\frac{2r+1}{2}x\right)}$$

$$g_n(x) = \prod_{r=1}^n f_r(x) \quad \forall n=1,2,3,\dots$$

5. If $I_n = \int_0^\pi \frac{f_n(x)}{g_n(x)} dx$ then $\sum_{K=1}^{100} I_K =$

- A) 25π B) 50π C) 75π D) 100π

6. The value of $\lim_{x \rightarrow 0} \int_0^x \frac{dt}{xf_9(t)g_9(t)} =$

- A) $\frac{50}{9}$ B) $\frac{10}{9}$ C) $\frac{100}{9}$ D) $\frac{25}{9}$

Paragraph-4

Let $f(x)$ be a polynomial of degree 4 satisfying

$$\left(\int_1^x A(t)B(t)dt\right) \cdot \left(\int_1^x C(t)D(t)dt\right) - \left(\int_1^x A(t)C(t)dt\right) \cdot \left(\int_1^x B(t)D(t)dt\right) = f(x), \forall x \in R$$

Where $A(x)$, $B(x)$, $C(x)$, $D(x)$ are non-constant continuous and differentiable functions. It is given that the leading coefficient of $f(x)$ is 1, then

7. Area included between the line $y = x-1$ and $y = f(x)$ is....

- A) $2/5$ B) $3/10$ C) $7/10$ D) $7/5$

8. Area of the smaller region intercepted between the curves $y = f(x)$ and $x^2 + y^2 = 1$, is...

- A) $\frac{\pi}{4} - \frac{1}{5}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{4} + \frac{1}{5}$ D) $\frac{\pi}{2} + \frac{1}{5}$

9. Area included between $y = f(x)$ and $y + 2 = 0$ between the lines $x = 0$ and $x = 3$, is.....

- A) $3/5$ B) $33/5$ C) $23/10$ D) $63/5$

SECTION-IV

[Matrix Matching Type]

1. Column – I

Column – II

- A) If $f(\pi) = 2$ and $\int_0^{\pi} (f(x) + f''(x)) \sin x \, dx = 5$, then p) 2
- $f(0)$ is equal to (It is given that $f(x)$ is continuous in $[0, \pi]$)
- B) The value of $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{1}{t(1+t^2)} dt$, where $x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ q) 1/2
- is equal to
- C) $\lim_{n \rightarrow \infty} \left(\sin \frac{\pi}{2n} \cdot \sin \frac{2\pi}{2n} \cdot \sin \frac{3\pi}{2n} \cdots \sin \frac{(n-1)\pi}{n} \right)^{1/n}$ is equal to r) 3
- D) If $f(x) = \begin{cases} 0 & , \text{ where } x = \frac{n}{n+1}, n=1,2,3,\dots \\ 1 & , \text{ else where} \end{cases}$, then the value of $\int_0^2 f(x) dx$ s) 1
- a) A – r ; B – s ; C – q ; D – p b) A – r ; B – s ; C – r ; D – q
- c) A – r ; B – s ; C – p ; D – q d) A – s ; B – r ; C – q ; D – p

2. Column – I

Column – II

- A) Let $f(x) = \text{minimum}(|x|, 1-|x|, 1/4)$, $\forall x \in \mathbb{R}$, then value of p) 59/6

$\int_{-1}^1 f(x) dx$ is equal to

- B) Let $f(x) = \int_0^x (t^2 - t + 1) dt$, $\forall x \in [3, 4]$, then the difference q) 1/14

between the greatest and the least values of the function is

- C) The integral $\int_{\pi/4}^{5\pi/4} (|\cos t| \sin t + |\sin t| \cos t) dt$ r) 0

has the value equal to

- D) $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r} + 4\sqrt{n})^2}$ is equal to s) 3/8

- a) A – s ; B – q ; C – r ; D – p b) A – s ; B – r ; C – p ; D – q
- c) A – s ; B – p ; C – r ; D – q d) A – p ; B – r ; C – s ; D – q

3. Match the following: $a < b; a, b \in \mathbb{N}$

Column I		Column II	
(A)	$\int_a^b \sqrt{\frac{x-a}{b-x}} dx$	(P)	$\frac{\pi}{\sqrt{ab}}$
(B)	$\int_a^b \frac{1}{\sqrt{(x-a)(b-x)}} dx$	(Q)	$(b-a)\pi/2$
(C)	$\int_a^b \frac{1}{x\sqrt{(x-a)(b-x)}} dx$	(R)	$(a+b)\pi/2$
(D)	$\int_a^b \frac{x}{\sqrt{(x-a)(b-x)}} dx$	(S)	π

4. Match the following:

Column -I

A) The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded the lines $x = 4, y = 4$ and the coordinate axes.

If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom, then

B) If S_1, S_2, S_3 are the areas bounded by the x-axis and any three consecutive half waves of the curves $y = e^{ax} \sin \beta x, x \geq 0$, from left to right, then $(a, \beta \neq 0)$

C) Let $f(x) = \max.\{x^2, (1-x)^2, 2x(1-x)\}$, where $0 \leq x \leq 1$ if the areas of the region

Column -II

(P) S_1, S_2, S_3 are in A.P.

(Q) S_1, S_2, S_3 are in G.P

(R) $S_1^3 + S_3^3 > 2S_2^3$

bounded by the curve $y = f(x)$ between

$$x \in \left[0, \frac{1}{3}\right], \left[\frac{1}{3}, \frac{2}{3}\right], \left[\frac{2}{3}, 1\right] \text{ and x-axis}$$

represented by S_1, S_2, S_3 then

D) If the area bounded by $y^2 = x, y = x$, (S) $S_1 = S_2 = S_3$

$x = \sqrt{y}, y = x$ and $y = x^2, y = x$ are denoted

By S_1, S_2 and S_3 respectively then

Yashpatil TG~ @bohring_bot

KEY

(MULTIPLE CORRECT ANSWER TYPE)

1.	AB	5.	AD	9.	A	13.	ACD	17.	AC	21.	ABC	25.	BCD
2.	ABCD	6.	ABCD	10.	AC	14.	AC	18.	BCD	22.	ABCD		
3.	A	7.	ABD	11.	ABCD	15.	AC	19.	BCD	23.	D		
4.	ABCD	8.	AC	12.	ABCD	16.	AD	20.	BCD	24.	ABCD		

(INTEGER ANSWER TYPE)

1.	0	5.	2	9.	7	13.	7	17.	2	21.	1	25.	6
2.	9	6.	5	10.	1	14.	3	18.	8	22.	5		
3.	2	7.	9	11.	8	15.	8	19.	4	23.	2		
4.	4	8.	9	12.	6	16.	4	20.	9	24.	6		

(PARAGRAPH TYPE)

1.	D	5.	B	9.	D
2.	C	6.	C		
3.	B	7.	B		
4.	A	8.	A		

[Matrix Matching Type]

1.	A	2.	C
3.	A-Q B-S C-P D-R	4.	A-PQS B-QR C-R D-PQS